Business Calculus **Name (Print):** ____________________________
MATH 2350

- This exam contains 14 pages (including the front and back of this cover page) and 26 problems. Check to see if any pages are missing.
- This is the first exam and will count for 20% of your final grade. You have 115 minutes (1:35 pm - 3:30 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- If a question is followed by $\boxed{\text{work}}$, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work when $\boxed{\textbf{work}}$ is required will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 350 points total in this exam, as well as 5 bonus questions worth a total of 45 possible points.

Page:	3	4	5	6	7	8	9	10	11	12	Total
Points:	60	35	30	15	15	40	45	50	50	10	350
Score:											

1. Determine where each function is continuous. Express the answer in interval notation.

   (a) $\boxed{\text{10 points}}$ $f(x) = \dfrac{x-2}{x^2+4}$

   $\boxed{\text{work}}$

   (b) $\boxed{\text{10 points}}$ $g(x) = (x+7)^1 8$

   $\boxed{\text{work}}$

2. Find each limit, if it exists.

   (a) $\boxed{\text{10 points}}$ $\displaystyle\lim_{x\to 0} \frac{x-7}{x^2-4x-21}$

   $\boxed{\text{work}}$

   (b) $\boxed{\text{10 points}}$ $\displaystyle\lim_{x\to 7} \frac{x-7}{x^2-4x-21}$

   $\boxed{\text{work}}$

   (c) $\boxed{\text{10 points}}$ $\displaystyle\lim_{x\to -3} \frac{x-7}{x^2-4x-21}$

   $\boxed{\text{work}}$

   (d) $\boxed{\text{10 points}}$ $\displaystyle\lim_{x\to 2} \frac{x-2}{|x-2|}$

   $\boxed{\text{work}}$

3. $\boxed{\text{10 points}}$ Let $\lim_{x\to 2} f(x) = \pi^2$ and $\lim_{x\to 2} g(x) = -9$. Find $\lim_{x\to 2} \dfrac{g(x) - 4\sqrt{f(x)}}{g(x) - f(x)}$.

$\boxed{\text{work}}$

4. $\boxed{\text{10 points}}$ Find $\lim_{h\to 0} \dfrac{f(7+h) - f(7)}{h}$ for $f(x) = -2x + 2$.

$\boxed{\text{work}}$

5. $\boxed{\text{15 points}}$ Find the average rate of change for the function $f(x) = 2x^2 + 7x + 1$ between $x = 1$ and $x = 5$.

$\boxed{\text{work}}$

6. $\boxed{\text{10 points}}$ Find the instantaneous rate of change for the function $f(x) = 2x^2 + 7x + 1$ at $x = 5$.

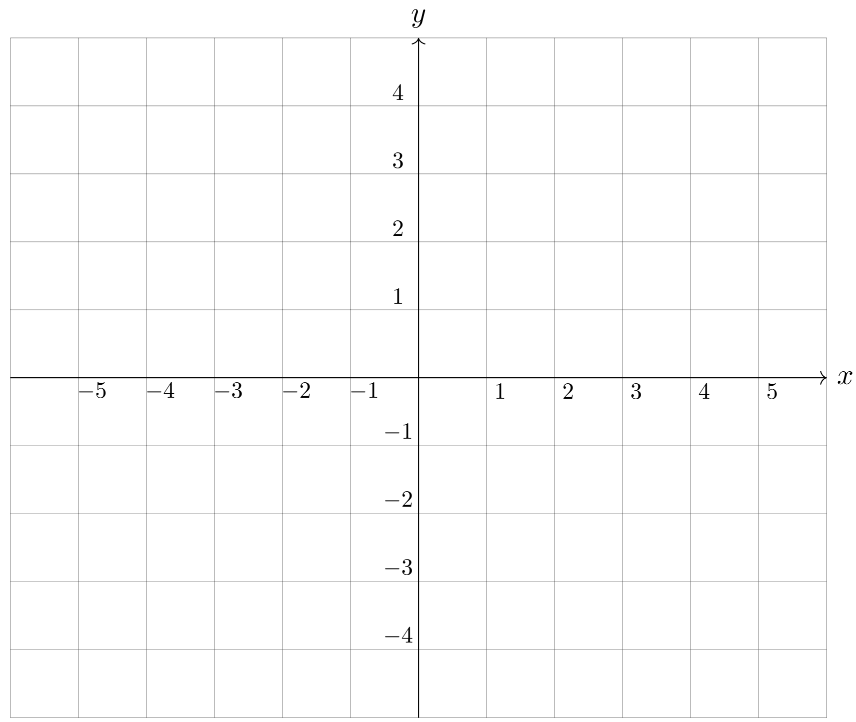
$\boxed{\text{work}}$

7. $\boxed{\text{20 points}}$ Using the definition of the derivative $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$, find the derivative of $f(x) = 2x^2 + 4$.

$\boxed{\text{work}}$

8. (a) $\boxed{\text{10 points}}$ Sketch a possible graph of a single function $f(x)$ that satisfies the given conditions. Assume that $f(x)$ is defined everywhere on $(-\infty, \infty)$.

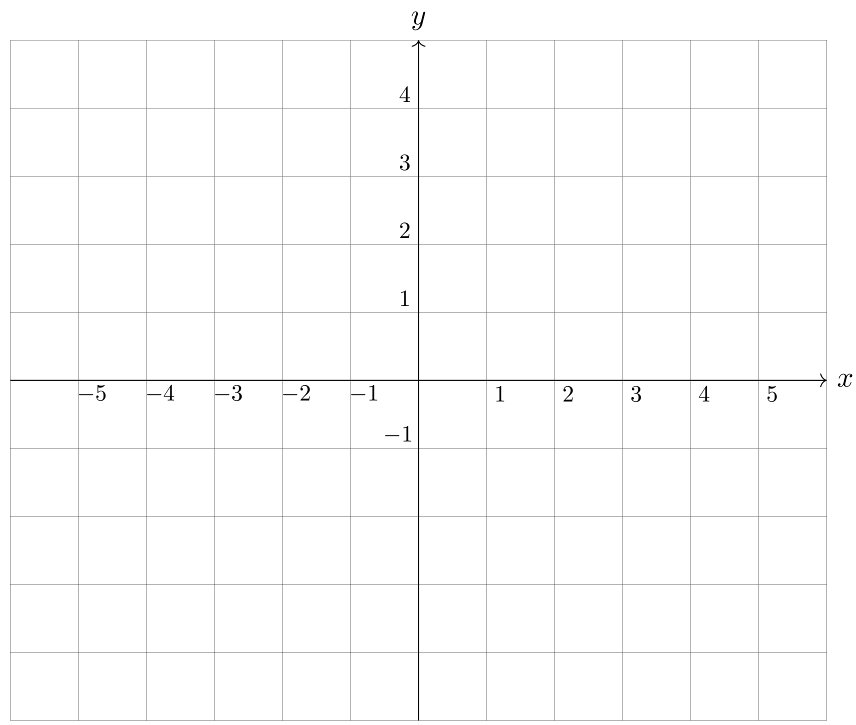
$$\lim_{x\to -3^-} f(x) = -1, \quad \lim_{x\to -3^+} f(x) = -3, \quad \lim_{x\to 0} f(x) = 0$$



(b) $\boxed{\text{5 points}}$ List all values of x where the function drawn above is discontinuous.

9. (a) $\boxed{\text{10 points}}$ Sketch a possible graph of a single function $f(x)$ that satisfies the given conditions. Assume that $f(x)$ is defined everywhere on $(-\infty, \infty)$.

$$\lim_{x \to 3^-} f(x) = 2, \quad \lim_{x \to 3^+} f(x) = 0, \quad \lim_{x \to 5} f(x) = -2$$



(b) $\boxed{\text{5 points}}$ List all values of x where the function drawn above is discontinuous.

10. Use the properties of derivatives and differentiation rules to find the indicated derivatives.

(a) $\boxed{\text{10 points}}$ $\dfrac{d}{dx}\left(3x^{-5} + x^4 + 3^5\right)$

$\boxed{\text{work}}$

(b) $\boxed{\text{10 points}}$ $f'(x)$, where $f(x) = \dfrac{x^4 - \sqrt[3]{x}}{2x}$

$\boxed{\text{work}}$

(c) $\boxed{\text{10 points}}$ $f'(x)$, where $f(x) = 7x^{5/7} - 5x^3 + 90000$

$\boxed{\text{work}}$

11. $\boxed{\text{10 points}}$ Let $f'(x) = x+3$ and $g'(x) = -3x^2+32$. Find $h'(x)$, where $h(x) = 3f(x)-g(x)+2$.

$\boxed{\text{work}}$

12. Let $F(x)$ be the function defined by $F(x) = \frac{1}{7}x^7 - 2x^6 + 13$.

(a) 10 points Find $F'(x)$.

work

(b) 20 points Determine the equation of the tangent line to $F(x)$ at $x = 1$.

work

(c) 15 points Specify the points at which the tangent line to $F(x)$ is flat (that is, where it has a slope of zero).

work

13. $\boxed{\text{10 points}}$ Let $f(x) = 4\left(e^x\right) - \ln(x^{100})$. Find $f'(x)$.

$\boxed{\text{work}}$

14. $\boxed{\text{10 points}}$ Compute $\dfrac{d}{dx}\left[3\ln(x) - \frac{1}{\pi}e^x\right]$.

$\boxed{\text{work}}$

15. $\boxed{\text{15 points}}$ Find y' for $y = \dfrac{5x-5}{x^2+2x}$.

$\boxed{\text{work}}$

16. $\boxed{\text{15 points}}$ Compute $\dfrac{d}{dt}\left[t(t-2)3e^t\right]$.

$\boxed{\text{work}}$

17. $\boxed{\text{15 points}}$ Find $f'(x)$ for $f(x) = 2\ln(x^5) - 3(\ln(x))^2$.

$\boxed{\text{work}}$

18. $\boxed{\text{15 points}}$ Compute $\dfrac{d}{dx}\left[\dfrac{x+1}{(x+3)^2}\right]$.

$\boxed{\text{work}}$

19. $\boxed{\text{10 points}}$ Suppose that the total profit in hundreds of dollars from selling x items is given by $P(x) = 2x^2 - 7x + 100$. Find the marginal profit at $x = 5$.

$\boxed{\text{work}}$

20. $\boxed{\text{10 points}}$ The revenue (in thousands of dollars) from producing x units of an item is modeled by $R(x) = 2x - 0.0002x^2$. Find the marginal revenue at $x = 500$.

$\boxed{\text{work}}$

21. 10 points | work — The total cost to produce x units of paint is $C(x) = (5x+3)(7x+4)$. Find the marginal average cost function.

22. 10 points (bonus) | work — You invest \$1,000 with a financial institution at an annual interest rate of 5% compounded continuously. Using the formula $A(t) = Pe^{rt}$, where $A(t)$ is the value of the investment after t years, r is the annual percentage rate, and P is the initial investment amount, find the instantaneous rate of change in the value of your investment with respect to time.

23. $\boxed{\text{5 points (bonus)}}$ $\boxed{\text{work}}$ Suppose the demand for a certain item is given by $D(p) = -2p^2 + 7p + 1$, where p represents the price of the item. Find $D'(p)$, the rate of change of demand with respect to price.

24. $\boxed{\text{10 points (bonus)}}$ $\boxed{\text{work}}$ Find $f'(x)$, where $f(x) = 5x^2e^x + 4ln(3x^3)$.

25. $\boxed{\text{10 points (bonus)}}$ $\boxed{\text{work}}$ Find the equation of the line tangent to the graph of the function

$$f(x) = 3 + 4xe^x \text{ at } x = 1.$$

26. $\boxed{\text{10 points (bonus)}}$ Find $f'(x)$, where $f(x) = (5x^3 + 4)(3x^7 - 5)$.

$\boxed{\text{work}}$